Regression Analysis and Callback Study

Causal Inference Assignment I

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1 Question 1: Treatment Effects Using the Lalonde Data

1.1 Data and Objective

The `cps1re74.csv` dataset is available in this repository under the **Causal Inference** folder. The objective is to estimate the effect of treatment status (`treat`) on real earnings in 1978 (`re78`). Two additional indicator variables are generated to identify individuals whose earnings were zero in 1974 or 1975.

The simple regression model is

$$\text{re78}_i = \beta_0 + \beta_1 \text{treat}_i + \varepsilon_i,$$

where β_1 represents the unadjusted difference in mean 1978 earnings between the treatment and comparison groups.

1.2 Load and Prepare the Data

```
library(tidyverse)

lalonde <- read.table(
  "C:/Users/USER/Downloads/cps1re74.csv",
  header = TRUE
) |>
  as_tibble() |>
  mutate(
    u74 = as.numeric(re74 == 0),
    u75 = as.numeric(re75 == 0)
  )

dim(lalonde)
```

```
[1] 16177    12
```

```
head(lalonde)
```

```
# A tibble: 6 x 12
  treat   age  educ black hispan married nodegree  re74  re75   re78   u74   u75
  <int> <int> <int> <int>  <int>   <int>    <int> <dbl> <dbl>  <dbl> <dbl> <dbl>
1     1    37    11     1      0       1        1     0     0  9930.     1     1
2     1    22     9     0      1       0        1     0     0  3596.     1     1
3     1    30    12     1      0       0        0     0     0 24909.     1     1
4     1    27    11     1      0       0        1     0     0  7506.     1     1
5     1    33     8     1      0       0        1     0     0   290.     1     1
6     1    22     9     1      0       0        1     0     0  4056.     1     1
```

The variables `u74` and `u75` equal 1 when earnings in the corresponding year are zero and 0 otherwise.

1.3 1(a) Regression of `re78` on `treat`

Estimate the simple linear regression model:

$$\text{re78}_i = \beta_0 + \beta_1 \text{treat}_i + \varepsilon_i.$$

```r
m1 <- lm(re78 ~ treat, data = lalonde)
summary(m1)
```

```
Call:
lm(formula = re78 ~ treat, data = lalonde)

Residuals:
   Min     1Q Median     3Q    Max
-14856  -9006   1444  10744  53959

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 14855.64      76.22  194.90   <2e-16 ***
treat       -8506.50     712.77  -11.93   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9639 on 16175 degrees of freedom
Multiple R-squared:  0.008729,  Adjusted R-squared:  0.008668
F-statistic: 142.4 on 1 and 16175 DF,  p-value: < 2.2e-16
```

The estimated coefficient of `treat` gives the estimated difference in mean 1978 earnings between treated and untreated individuals. Describe the coefficient, its statistical significance, and the model's R^2.

1.4 1(b) Regression of `re78` on All Available Predictors

Fit a multiple linear regression model including `treat` and all other available predictors, including the two generated indicators.

$$\text{re78}_i = \beta_0 + \beta_1 \text{treat}_i + \sum_{j=2}^{p} \beta_j X_{ij} + \varepsilon_i,$$

where X_{ij} denotes the additional predictors.

```r
m2 <- lm(re78 ~ ., data = lalonde)
summary(m2)
```

```
Call:
lm(formula = re78 ~ ., data = lalonde)

Residuals:
   Min     1Q Median     3Q    Max
-25032  -3524   1294   3774  53742

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.762e+03  4.456e+02  12.931  < 2e-16 ***
treat        1.068e+03  5.541e+02   1.927 0.054026 .
age         -9.454e+01  6.000e+00 -15.756  < 2e-16 ***
educ         1.752e+02  2.870e+01   6.106 1.04e-09 ***
black       -8.111e+02  2.128e+02  -3.811 0.000139 ***
hispan      -2.305e+02  2.186e+02  -1.055 0.291647
married      1.532e+02  1.428e+02   1.073 0.283190
nodegree     3.429e+02  1.779e+02   1.928 0.053888 .
re74         2.914e-01  1.273e-02  22.891  < 2e-16 ***
re75         4.427e-01  1.289e-02  34.353  < 2e-16 ***
u74          3.556e+02  2.316e+02   1.535 0.124751
u75         -1.613e+03  2.398e+02  -6.725 1.81e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6999 on 16165 degrees of freedom
Multiple R-squared:  0.4777,    Adjusted R-squared:  0.4773
F-statistic:  1344 on 11 and 16165 DF,  p-value: < 2.2e-16
```

In this model, the coefficient of `treat` estimates the difference in 1978 earnings between treatment groups after adjusting for the other included predictors.

1.5 1(c) Regression Models Using All Subsets of Predictors

Fit every possible subset of the ten candidate predictors while including `treat` in every model.

The candidate predictors are:

```r
x <- c(
  "age", "educ", "black", "hispan", "married", "nodegree",
  "re74", "re75", "u74", "u75"
)
```

For each subset S, the regression model is

$$\text{re78}_i = \beta_0 + \beta_1 \text{treat}_i + \sum_{j \in S} \beta_j X_{ij} + \varepsilon_i.$$

1.5.1 Fit All Subset Models

```
y <- "re78"

combs <- unlist(
  lapply(
    0:length(x),
    function(k) combn(x, k, simplify = FALSE)
  ),
  recursive = FALSE
)

res <- data.frame(
  regs = character(),
  coef = numeric(),
  p_value = numeric(),
  direction = character(),
  significance = character(),
  stringsAsFactors = FALSE
)

for (j in combs) {

  rhs <- c("treat", j)

  formula <- as.formula(
    paste(y, "~", paste(rhs, collapse = " + "))
  )

  fit <- lm(formula, data = lalonde)
  coef_table <- summary(fit)$coefficients

  treatment_estimate <- coef_table["treat", "Estimate"]
  treatment_p_value <- coef_table["treat", "Pr(>|t|)"]

  res <- rbind(
    res,
    data.frame(
      regs = if (length(j) == 0) {
        "(No additional predictors)"
      } else {
        paste(j, collapse = ", ")
      },
      coef = treatment_estimate,
      p_value = treatment_p_value,
      direction = ifelse(
        treatment_estimate > 0,
```

```
        "Positive",
        "Negative"
      ),
      significance = ifelse(
        treatment_p_value < 0.05,
        "Significant",
        "Not significant"
      ),
      stringsAsFactors = FALSE
    )
  )
}

head(res)
```

```
                        regs      coef      p_value direction significance
1 (No additional predictors) -8506.495 1.073367e-32  Negative  Significant
2                        age -7507.724 2.561893e-26  Negative  Significant
3                       educ -7744.668 8.916655e-28  Negative  Significant
4                      black -6172.819 1.163855e-16  Negative  Significant
5                     hispan -8525.889 6.910967e-33  Negative  Significant
6                    married -5719.118 2.088426e-16  Negative  Significant
```

There are $2^{10} = 1024$ possible subsets of the ten candidate predictors. Thus, 1,024 regression models are fitted, each including `treat`.

1.5.2 Count Significant and Nonsignificant Coefficients

```
table(
  factor(
    res$significance,
    levels = c("Significant", "Not significant")
  )
)
```

```
    Significant Not significant
            340             684
```

1.5.3 Count Positive and Negative Significant Coefficients

```r
significant_direction_counts <- table(
  factor(
    res$direction,
    levels = c("Positive", "Negative")
  ),
  factor(
    res$significance,
    levels = c("Significant", "Not significant")
  )
)

significant_direction_counts
```

```
          Significant Not significant
  Positive         260             549
  Negative          80             135
```

```r
# Positively significant coefficients
significant_direction_counts["Positive", "Significant"]
```

```
[1] 260
```

```r
# Negatively significant coefficients
significant_direction_counts["Negative", "Significant"]
```

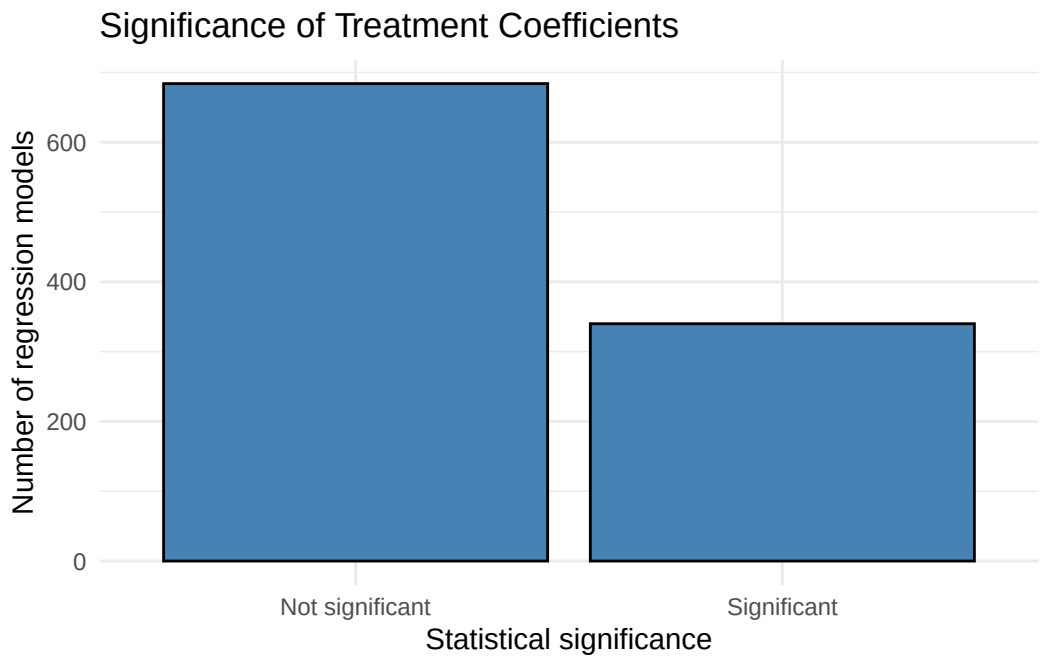
```
[1] 80
```

```r
# Nonsignificant coefficients
sum(res$significance == "Not significant")
```

```
[1] 684
```

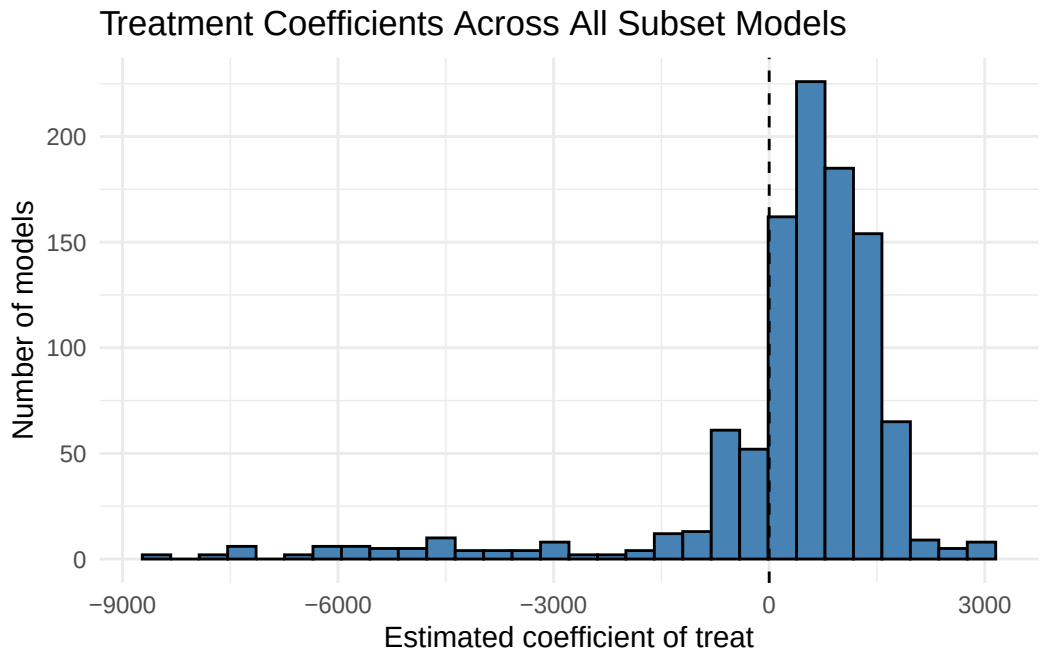
1.5.4 Visualize Statistical Significance

```r
ggplot(res, aes(x = significance)) +
  geom_bar(fill = "steelblue", color = "black") +
  labs(
    title = "Significance of Treatment Coefficients",
    x = "Statistical significance",
    y = "Number of regression models"
  ) +
  theme_minimal()
```



1.5.5 Distribution of Treatment Coefficients

```
ggplot(res, aes(x = coef)) +
  geom_histogram(
    bins = 30,
    fill = "steelblue",
    color = "black"
  ) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Treatment Coefficients Across All Subset Models",
    x = "Estimated coefficient of treat",
    y = "Number of models"
  ) +
  theme_minimal()
```



1.5.6 Models with the Smallest P-values

```
res |>
  arrange(p_value) |>
  slice_head(n = 10)
```

```
                        regs      coef      p_value direction significance
1                     hispan -8525.889 6.910967e-33  Negative  Significant
2  (No additional predictors) -8506.495 1.073367e-32  Negative  Significant
3                 educ, hispan -7770.060 6.049030e-28  Negative  Significant
4                         educ -7744.668 8.916655e-28  Negative  Significant
5                  age, hispan -7528.022 1.721446e-26  Negative  Significant
6                          age -7507.724 2.561893e-26  Negative  Significant
7       educ, hispan, nodegree -7243.334 1.575990e-24  Negative  Significant
8                educ, nodegree -7218.201 2.250558e-24  Negative  Significant
9             hispan, nodegree -7218.290 2.385561e-24  Negative  Significant
10                    nodegree -7184.181 3.891774e-24  Negative  Significant
```

2 Question 2: Race and Callback Rates in the Resume Data

2.1 Data and Objective

The `resume.csv` dataset is provided through Google Classroom. The objective is to examine whether race is associated with callback status. The variable `race` identifies the race group, while `call` indicates whether an applicant received a callback.

2.2 Load the Data

```
resume <- read_csv("C:/Users/USER/Downloads/resume.csv")

head(resume)
```

```
# A tibble: 6 x 4
  firstname sex    race   call
  <chr>     <chr>  <chr> <dbl>
1 Allison   female white     0
2 Kristen   female white     0
3 Lakisha   female black     0
4 Latonya   female black     0
5 Carrie    female white     0
6 Jay       male   white     0
```

```
glimpse(resume)
```

```
Rows: 4,870
Columns: 4
$ firstname <chr> "Allison", "Kristen", "Lakisha", "Latonya", "Carrie", "Jay",~
$ sex       <chr> "female", "female", "female", "female", "female", "male", "f~
$ race      <chr> "white", "white", "black", "black", "white", "white", "white~
$ call      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
```

2.3 2(a) Summary Table by Race

Calculate the total number of applicants, numbers of male and female applicants, number of callbacks, and callback rate for each race group.

```
race_summary <- resume |>
  group_by(race) |>
  summarise(
    total = n(),
    male = sum(sex == "male"),
    female = sum(sex == "female"),
    callback = sum(call == 1),
    callback_rate = mean(call),
    .groups = "drop"
  )

race_summary
```

```
# A tibble: 2 x 6
  race  total  male female callback callback_rate
  <chr> <int> <int>  <int>    <int>         <dbl>
1 black  2435   549   1886      157        0.0645
2 white  2435   575   1860      235        0.0965
```

The callback rate is calculated as

$$\hat{p}_{\text{callback}} = \frac{\text{Number of applicants receiving callbacks}}{\text{Total number of applicants}}.$$

2.4 2(b) Contingency Table and Chi-squared Test

Construct a contingency table of race and callback status.

```
tab <- table(resume$race, resume$call)

tab
```

```
          0   1
  black 2278 157
  white 2200 235
```

The hypotheses are

$$H_0 : \text{Race and callback status are independent,}$$

$$H_1 : \text{Race and callback status are associated.}$$

The Pearson chi-squared test statistic is

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

where O_{ij} and E_{ij} are the observed and expected cell counts.

```
chisq.test(tab)
```

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  tab
X-squared = 16.449, df = 1, p-value = 4.998e-05
```

At the 5% significance level, compare the p-value with 0.05 to determine whether there is evidence of an association between race and callback status.

2.5 2(c) Subgroup Analyses by Sex

Repeat the contingency-table analysis separately for male and female applicants.

2.5.1 Male Applicants

```
male_data <- resume |>
  filter(sex == "male")

tab_m <- table(male_data$race, male_data$call)

tab_m
```

```
         0   1
  black 517  32
  white 524  51
```

```
chisq.test(tab_m)
```

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  tab_m
X-squared = 3.3655, df = 1, p-value = 0.06658
```

2.5.2 Female Applicants

```
female_data <- resume |>
  filter(sex == "female")

tab_f <- table(female_data$race, female_data$call)

tab_f
```

```
           0    1
  black 1761  125
  white 1676  184
```

```r
chisq.test(tab_f)
```

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  tab_f
X-squared = 12.76, df = 1, p-value = 0.0003541
```

Compare the p-values and callback patterns from the two subgroup analyses. Discuss whether the evidence of an association between race and callback status appears similar for male and female applicants.